

# JEE MAIN REVISION MASTER: LOGARITHM

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                           | Frequency | Category              |
|---------------|--------------------------------------------------------|-----------|-----------------------|
| Moderate      | Logarithm — Core Principles & Governing Laws           | 6 PYQs    | Core Principle        |
| Standard      | Logarithm — Standard Calculations & Problem Archetypes | 2 PYQs    | Quantitative Analysis |
| Standard      | Logarithm — Applied Cases, Reactions & Traps           | 0 PYQs    | Applied Topics        |

## 2. Master Formula Reference Card

| Concept / Topic                                        | Mathematical Formulation & Governing Equations                                                                                                                    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Logarithm — Core Principles & Governing Laws           | $e^{5(\log_e x)^2 + 3} = x^8, x > 0, \quad \log_{\frac{1}{2}}  \sin x , \quad 1 = \log_2 (x$                                                                      |
| Logarithm — Standard Calculations & Problem Archetypes | $x > -\frac{5}{2}, \quad \beta = \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots \infty, \quad \alpha = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots \infty$ |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Logarithm — Core Principles & Governing Laws (Moderate Yield — 6 PYQs)

**Core Principle:** Fundamental definitions, core postulates, and governing physical/chemical relations in Logarithm.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Logarithm.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Logarithm PYQs.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#2 (MCQ): "The sum of all the real solutions of the equation  $\log_{((x+3))(6x^2+28x+30)} = 5 - 2\log_{((6x+10))(x^2+6x+9)}$  is equal to :..." [Ans: N/A]
- [2025 Morning Shift] Q#3 (MCQ): "The product of all solutions of the equation  $e^{5(\log_e x)^2 + 3} = x^8, x > 0$ , is :..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "If the solution of the equation  $\log_{(\cos x)} \cot x + 4\log_{(\sin x)} \tan x = 1, x \in (0, (\pi / 2))$ , is  $\sin^{-1}(\frac{\alpha}{\sqrt{\beta}})$ , ..." [Ans: N/A]

### 3.2 Logarithm — Standard Calculations & Problem Archetypes (Standard Yield — 2 PYQs)

**Core Principle:** Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Logarithm.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "Let  $\alpha = (1/4) + (1/8) + (1/16) + \dots$  and  $\beta = (1/3) + (1/9) + (1/27) + \dots$ . Then the value ..." [Ans: C]
- [2023 Morning Shift] Q#4 (MCQ): "The number of integral solutions  $x$  of  $\log_{((x+(7/2)))}((x-7/2x-3))^2 \geq 0$  is :..." [Ans: N/A]

### 3.3 Logarithm — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

**Core Principle:** Special cases, experimental observations, and conceptual traps designed to test deep understanding of Logarithm.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

**Representative Source PYQs:**

- [2026 Evening Shift] Q#1 (MCQ): "Let  $\alpha = (1/4) + (1/8) + (1/16) + \dots$  and  $\beta = (1/3) + (1/9) + (1/27) + \dots$ . Then the value ..." [Ans: C]
- [2026 Evening Shift] Q#2 (MCQ): "The sum of all the real solutions of the equation  $\log_{((x+3))}(6x^2+28x+30) = 5 - 2\log_{((6x+10))}(x^2+6x+9)$  is equal to :..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

Let  $\alpha = (1/4) + (1/8) + (1/16) + \dots$  and  $\beta = (1/3) + (1/9) + (1/27) + \dots$ . Then the value of  $(0.2)^{\log_{(\sqrt{5})}(\alpha)} + (0.04)^{\log_{(5)}(\beta)}$  is equal to :

- (A) 4
- (B) 5
- (C) 8 [Correct]
- (D) 25

**Correct Answer: C**

**Solution Analysis:** We first find the values of  $\alpha$  and  $\beta$ . Given  $\alpha = (1/4) + (1/8) + (1/16) + \dots$ . This is an infinite geometric progression with  $a = (1/4)$ ,  $r = (1/2)$ . So,  $\alpha = (a / (1-r)) = ((1/4) / (1-(1/2))) = ((1/4) / (1/2)) = (1/2)$ . Now,  $\beta = (1/3) + (1/9) + (1/27) + \dots$ . This is also an infinite geometric progression with  $a = (1/3)$ ,  $r = (1/3)$ . Thus,  $\beta = (a / (1-r)) = ((1/3) / (1-(1/3))) = ((1/3) / (2/3)) = (1/2)$ . So both are equal to  $(1/2)$ . Now the expression becomes

**Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Evening Shift (Q#2)**

The sum of all the real solutions of the equation  $\log_{((x+3))}(6x^2+28x+30)=5-2\log_{((6x+10))}(x^2+6x+9)$  is equal to :

- (A) 1
- (B) 4
- (C) 0
- (D) 2

**Correct Answer: None**

**Solution Analysis:**  $\log_{((x+3))}(6x^2+28x+30)=5-2\log_{((6x+10))}(x^2+6x+9)-1$

$$\log_{((x+3))}2(3x^2+14x+15)+2\log_{((6x+10))}(x^2+6x+9)=5 \quad \log_{((x+3))}2(x+3)(3x+5)+2\log_{((6x+10))}(x+3)^2=5$$

$$\log_{((x+3))}(x+3)+\log_{((x+3))}(6x+10)+4\log_{((6x+10))}(x+3)=5 \quad \log_{((x+3))}(6x+10)+4\log_{((6x+10))}(x+3)=4\log_{((x+3))}(x+3)=1$$

Let,  $\log_{((x+3))}(6x+10)=t \Rightarrow \log_{((6x+10))}(x+3)=(1/t) \therefore t+4 \times (1/t)=4 \quad t^2-4t+4=0 \quad (t-2)^2=0 \quad \blacksquare t=2$   $\begin{aligned} & \text{\texttt{Checking}} x=1 \text{\texttt{in equation (1)}} \quad \& \log_{((1+3))}(6(1)) \end{aligned}$

$$\log_{((1+3))}(6(1))=2$$